10. $(q \wedge s) \wedge \neg(q \wedge s)$ T, 8, 9 and Conjunction

11. F T, 10 and negation law

Hence the given premises are inconsistent.

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MODULE-3

- **Set**

→ It is an unordered collection of objects

→ $a \in A$ 'a' is an element of set A.

→ Repetation of elements not allowed.

→ N = natural numbers = $\{0, 1, 2, 3, \ldots\}$ → whole numbers

→ Z = Integers = $\{\ldots, -3, -2, -1, 0, 1, 2, 3, \ldots\}$

→ Z^+ = positive integers = $\{0, 1, 2, 3, \ldots\}$

→ R = set of real numbers

→ R^+ = set of all positive real numbers

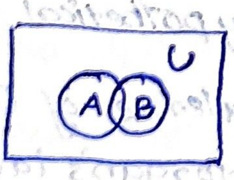
→ C = Complex numbers

→ Q = Rational numbes $\{x; p/q$, $q \neq 0$, p & q are positive integers

→ Intervals :- closed interval [ ] – included

open interval ( ) not included

→ Venn diagram



→ equal sets

$\forall x (x \in A \leftrightarrow x \in B)$

$\{1,2,3\} = \{2,1,3\}$

$\{1,1,2,3,4\} = \{1,2,3,4\}$

→ subsets

$A \subseteq B$ holds iff $\forall x (x \in A \rightarrow x \in B)$ is true

→ $\forall x (x \in A \leftrightarrow x \in B)$

$\forall x (x \in A \rightarrow x \in B) \wedge \forall x (x \in B \rightarrow x \in A)$

$\Rightarrow$ $A \subseteq B$ and $B \subseteq A$

→ Proper Subsets ($A \subset B$)

$\forall x (x \in A \rightarrow x \in B) \wedge \exists x (x \in B \wedge x \notin A)$

→ $|A|$: the Cardinality of set A

- The no. of distinct elements in A

Eg:- $|\emptyset| = 0$

$|\{1,2,3\}| = 3$

$|\{\phi\}| = 1$

~~Cardinality~~ No. of subsets $= 2^n$ $\quad n \rightarrow$ no. of elements

→ Power sets ($P(A)$)

$A = \{a, b\}$ then,

$P(A) = \{\{\phi\}, \{a\}, \{b\}, \{a,b\}\}$

→ **Tuples**

- 2-tuples are called ordered pairs.
- Eg:-) $(a,b,c) \neq (a,c,b)$

→ **Cartesian product**

$$A \times B = \{(a,b) : a \in A \wedge b \in B\}$$

Eg:- $A = \{a,b\}$ $B = \{1,2,3\}$

$$A \times B = \{(a,1),(a,2),(a,3),(b,1),(b,2),(b,3)\}$$

No. of ordered pairs $= A * B = \underline{\underline{6}}$

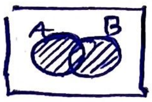
→ Relation is a subset of Cartesian product

→ ~~Truth set of~~ **Union** $(\cup)$

$\{1,2,3\} \cup \{3,4,5\}$

$\{x : x \in A \vee x \in B\}$

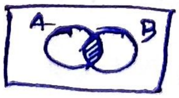
$\{1,2,3,4,5\}$



→ **Intersection** $(\cap)$

$\{1,2,3\} \cap \{3,4,5\}$

$\{x : x \in A \wedge x \in B\}$

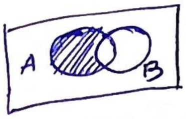


$\{3\}$

→ **Compliment** $(A' = U - A)$

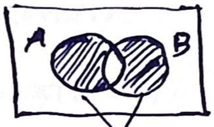
$\{x \in U : x \notin A\}$

→ A − B (Difference)



→ Symmetric Difference ($A \oplus B$)

$A \oplus B = (A-B) \cup (B-A)$



Eg:- $A = \{1, 2, 3, 4, 5\}$

$B = \{4, 5, 6, 7, 8\}$

$A \oplus B = (A-B) \cup (B-A)$

$(A-B) = \{$

→ Identity laws

$A \cup \phi = A \qquad A \cap U = A$

→ Domination laws

$A \cup U = U \qquad A \cap \phi = \phi$

→ $A \cup$

→ De Morgan's law

$\overline{A \cup B} = \bar{A} \cap \bar{B}$

$\overline{A \cap B} = \bar{A} \cup \bar{B}$

→ Absorption laws

$A \cup (A \cap B) = A$

• Proof of Second De Morgan Law

→ $\overline{A \cap B} \subseteq \bar{A} \cup \bar{B}$

ans:) $x \in \overline{A \cap B}$ by assumption

$x \notin A \cap B$ defn. of compliment

$\neg((x \in A) \wedge (x \in B))$ defn of intersection

$\neg(x \in A) \vee \neg(x \in B)$ 1^{st} De Morgan Law for Prop Logic

$x \notin A \vee x \notin B$ defn. of negation

$x \in \bar{A} \vee x \in \bar{B}$ defn of complement

$x \in \bar{A} \cup \bar{B}$ defn of union.

→ $\bar{A} \cup \bar{B} \subseteq \overline{A \cap B}$

ans:) $x \in \bar{A} \cup \bar{B}$ by assumption

$(x \in \bar{A}) \vee (x \in \bar{B})$ defn of union

$(x \notin A) \vee (x \notin B)$ defn of complement

$\neg(x \in A) \vee \neg(x \in B)$ defn of negation

$\neg((x \in A) \wedge (x \in B))$ by 1st De Morgan law for prop logic

$\neg(x \in A \cap B)$ defn of intersection

$x \in \overline{A \cap B}$ defn of compliment.

→ Using Membership table

$A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

A	B	C	$B \cap C$	$A \cup (B \cap C)$	$A \cup B$	$A \cup C$	$(A \cup B) \cap (A \cup C)$
1	1	1	1	1	1	1	1
1	1	0	0	1	1	1	1
1	0	1	0	1	1	1	1
1	0	0	0	1	1	1	1
0	1	1	1	1	1	1	1
0	1	0	0	0	1	0	0
0	0	1	0	0	0	1	0
0	0	0	0	0	0	0	0

$\rightarrow x \in A \cup (B \cap C)$

$x \in A \vee x \in (B \cap C)$

$x \in A \vee (x \in B \wedge x \in C)$

$x \in A \vee x \in B \wedge x \in A \vee x \in C$

$x \in A \cup B \wedge x \in A \cup C$

$x \in (A \cup B) \cap (A \cup C)$

$\rightarrow x \in (A \cup B) \cap (A \cup C)$

$x \in A \cup B \wedge x \in A \cup C$

$x \in A \vee x \in B \wedge x \in A \vee x \in C$

$x \in A \vee (x \in B \wedge x \in C)$

$x \in A \vee (x \in (B \cap C))$

$x \in A \cup (B \cap C)$

17/2/24 • <u>Binary relations</u>

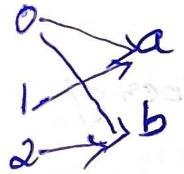
Binary Relation R from a set A to a set B is a subset

$R \subseteq A \times B$

Eg:-) Let $A = \{0,1,2\}$ and $B = \{a,b\}$

$\{(0,a),(0,b),(1,a),\cancel{(1,b)},(2,b)\}$ is a relation from A to B.

we can represent relations from a set A to a set B graphically or using a table:



R	a	b
0	x	x
1	x	
2		x

Note: → No. of relations in a set of 'n' elements = 2^{n^2}

→ Empty relation on an empty set is reflexive.

Antisymmetric Relation.

check.

Eg:- $\{1, 2, 3\}$ $\{(0,1), (2,3), (4,5)\}$ ↳ antisymmetric.

$\{(1,2), (2,1), (3,2)\}$ ↳ symmetric. $\{\cancel{(2,2)}, \cancel{(2,3)}, \cancel{(3,2)}\}$

If. a,b & b,a exist then $a=b$ ⇒ antisymmetric.

If. a,b & b,a exist ⇒ Symmetric. If there is nothing to compare check

Eg:-) $R = \{(a,b) : a = b+1\}$ → antisymmetric.

$\{(1,0), (2,1), (2,3)\}$

Reflexive relation.

check

Eg:- $\{1, 2, 3\}$

$\{(1,1), (2,2), (3,3)\}$

Transitive relation

Eg:- $\{1, 2, 3\}$

$\{(1,2), (2,3), (1,3)\}$

a,b & b,c

then (a,c) should exist.

Q) $R = \{(a,b) : a = b \text{ or } a = -b\}$

an) $\{(1,1), (2,2), (1,-1), (-1,1)\}$ ⇒ transitive.

Composition

If (x, y) is a member of R_1 and (y, z) is a member of R_2 then (x, z) is a member of $R_2 \circ R_1$

Equivalence relation

→ If it is symmetric, reflexive & transitive then it is an equivalence relation.

→ Two elements a, b are related by an equivalence relation is called an equivalent

$a \sim b$

Congruence Modulo m

Let m be an integer with $m > 1$.

Show that the relation $R = \{(a, b) : a \equiv b \pmod{m}\}$ is an equivalence relation on the set of integers.

Sol: Recall that $a \equiv b \pmod{m}$ if and only if m divides $a - b$.

Eg:- $17 \equiv 5 \pmod{6}$ because $17 - 5 = 12$ is divisible by 6

$20 \equiv 0 \pmod{5}$ since 20 is itself a multiple of 5

$20 \div 5 =$ remainder $= 0$

$0 \div 5 =$ remainder $= 0$

also $20 - 0 = 20$

$20 \div 5$: is divisible

Properties

→ Reflexive : $a \equiv a \pmod m$

→ Symmetric : If $a \equiv b \pmod m$, then $b \equiv a \pmod m$

→ Transitive : If $a \equiv b \pmod m$ and $b \equiv c \pmod m$, then $a \equiv c \pmod m$

• Functions

→ A function from A to B, denoted $f: A \to B$ is an assignment of each element of A to exactly one element of B.

→ Every relation is not a function but every function is a relation.

→ A function $f: A \to B$ is a subset of $A \times B$ (a relation)

→ Restriction: A relation where no two elements of the

rela

•) Partial ordering or partial order relation

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→ If R is reflexive, antisymmetric & transitive then it is partial order relation.

→ A set A together with a partial order relation R is called partially ordered set or poset.

$(\mathbb{Z}, \geq)$ Integers greater than or equal to is a poset

$aRb \mid a \geq b$

• Equivalence Classes

If R is an equivalence relation on a set A. The set of all elements of A that are related to an element 'a' of 'A' is called equivalence class of A and is denoted by '[a]$_R$'

Eg:- $A = \{1,2,3\}$

$R = \{(1,1), (1,2), (2,1), (2,2), (3,3)\}$

$[1]_R = \{1,2\}$

$[2]_R = \{1,2\}$ only the y part

$[3]_R = \{3\}$

• ~~Parse~~ Quotient Set of A

The collection of all equivalence classes of elements of A, under an equivalent relation R is denoted by A/R and is called quotient set of A by R and it is ~~denote~~ defined as

$$A/R = \{[a] / a \in A\}$$

Eg:- $A = \{1,2,3\}$

$R = \{(1,1), (1,2), (2,1), (2,2), (3,3)\}$

$[1]_R = \{1,2\}$

$[2]_R = \{1,2\}$

$[3]_R = \{3\}$

$A/R = \{\{1,2\}, \{3\}\}$

no need of repetition in a set

• Partition of a set

If 'S' is a non-empty set, a collection of disjoint non-empty subsets of 'S' whose union is 'S' is called a partition of 'S'.

Eg :- $S = \{1, 2, 3, 4\}$

$S_1 = \{1, 2\}$ $S_2 = \{3, 4\}$ ✓ $S_1 \cap S_2 = \phi$ ✓ $S_1 \cup S_2 = \{1, 2, 3, 4\} = S$

$S_3 = \{1, 2, 3\}$ $S_4 = \phi$ $S_4 = \times$ (empty set)

$S_3 \cap S_2 \neq \phi$ × $S_3 \cap S_2 \neq \phi$ ×

subsets in a partition is called block of partition

Eg:- $A = \{a_1, a_2, a_3\}$

$B = \{b_1, b_2, b_3, b_4\}$

$R = \{(a_1, b_2), (a_2, b_1), (a_2, b_3), (a_2, b_4), (a_3, b_2), (a_3, b_4)\}$

$R : A \rightarrow B$

	b_1	b_2	b_3	b_4
a_1	0	1	0	0
a_2	1	0	1	1
a_3	0	1	0	1

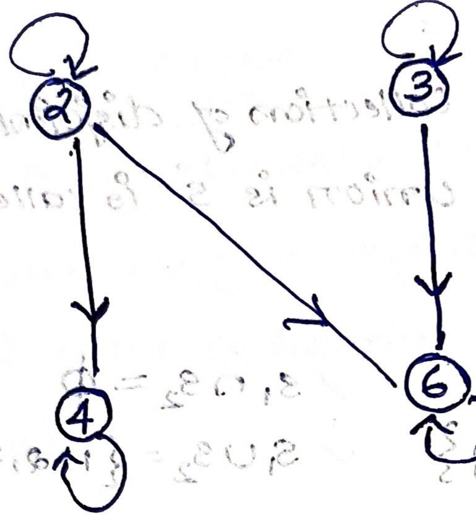
→ matrix representation.

Representation of a relation by using graph.

$A = \{2, 3, 4, 6\}$

R is defined by a R b if a/b   (a related to b; a divides b)

$R = \{(2,4), (3,6), (2,2), (3,3), (4,4), (6,6), (2,6)\}$

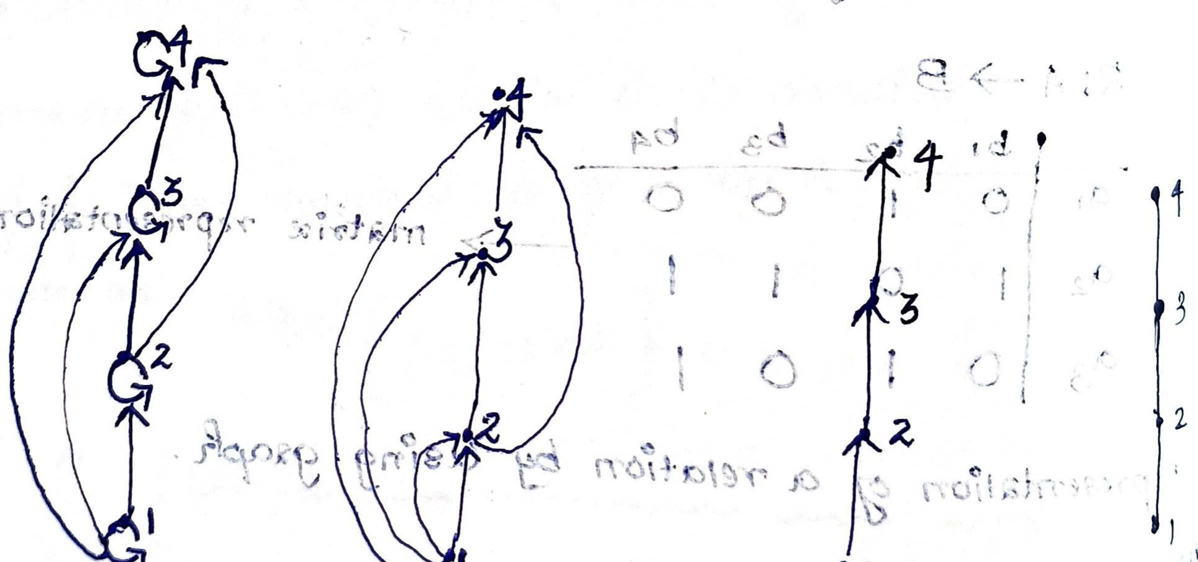


Digraph or Directed graph.
↓
Only possible for a relation defined for that set itself.

v.imp
- <u>Hasse Diagram for partial orderings</u>

Construct Hasse diagram for the partial ordering $\{(a,b) : a \leq b\}$ on a set $\{1, 2, 3, 4\}$

ans:) $R = \{(1,1), (2,2), (3,3), (4,4), (1,2), (1,3), (1,4), (2,3), (2,4), (3,4)\}$



by removing repetition (1,1), (2,2) ← by removing transitive relation (1,2), (2,3) then (1,3)

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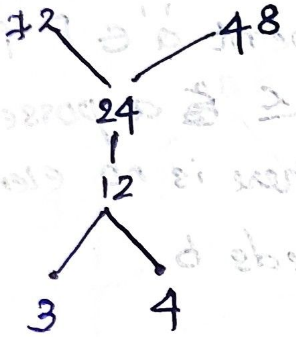
•) $A = (\{3, 4, 12, 24, 48, 72\}, \mid)$ ← a/b a divides b.

ans) $R = \{(3,12), (3,24), (3,48), (3,72), (4,12), (4,24), (4,48), (4,72), (12,24), (12,48), (12,72), (24,48), (24,72)\}$



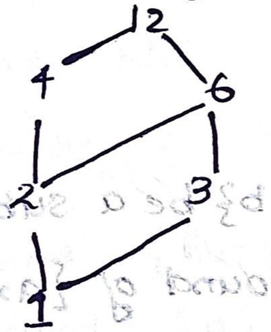
Maximal element = 72, 48

Minimal element = 3, 4

No greatest & least element.

•) $(D_{12}, \mid)$ set of positive integer divisors of 12 → D_{12}
↳ divides

ans; $\{1, 2, 3, 4, 6, 12\}$



$R = \{(1,2), (1,3), (1,4), (1,6), (1,12), (2,4), (2,6), (2,12), (3,6), (3,12), (4,12)\}$

Maximal element = 12

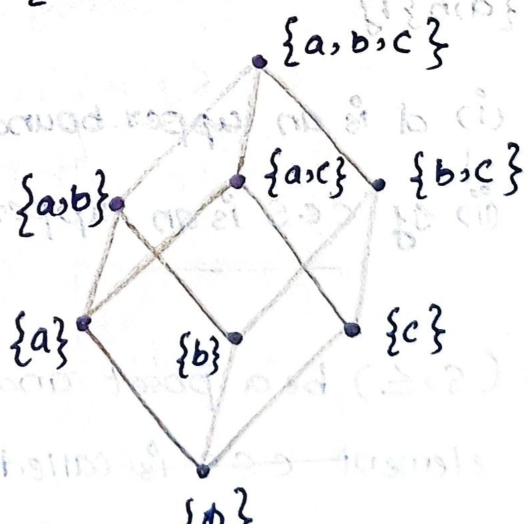
Minimal element = 1

greatest element = 12

Least element = 1

•) $(\{a, b, c\}, \leq)$ $2^3 = 8$.

ans:) $\phi, \{a\}, \{b\}, \{c\}, \{a,b\}, \{b,c\}, \{a,c\}, \{a,b,c\}$



Maximal element = $\{a, b, c\}$ → also the greatest element

Minimal element = $\{\phi\}$ → also the least element

- $(a,b) \in R$, R is partial order relation ($a \preceq b$) can be read as a preceeds b, b succeeds a

- **Maximal member :-**

When $(P, \preceq)$ is a poset an ~~at~~ element $a \in P$ is called a maximal member of P when $P, \preceq$ is a poset is called a maximal member of P, if there is no element $b \in P$ such that a strictly ~~br~~ preceeds b.

- If there exist an element $a \in P$ such that $b \preceq a$ (b preceeds a) for all $b \in P$, then a is called greatest member of the poset.

- **Partially ordered sets**

Let $(S, \leq)$ be a poset and let $\{a, b\}$ be a subset of S. An element $c \in S$ is called an upper bound of $\{a, b\}$ if $a \leq c$ and $b \leq c$.

An element $d \in S$ is called a least upper bound (lub) of $\{a,b\}$ if

(i) d is an upper bound of $\{a,b\}$; and

(ii) If $c \in S$ is an upper bound of $\{a,b\}$, then $d \leq c$.

Let $(S, \leq)$ be a poset and let $\{a,b\}$ be a subset of S. An element $c \in S$ is called a lower bound of $\{a,b\}$ if $c \leq a$ and $c \leq b$.

An element $d \in S$ is called a greatest lower bound (glb) of $\{a,b\}$ if

i) d is a lower bound of $\{a,b\}$ and

ii) If $c \in S$ is a lower bound of $\{a,b\}$ then $c \leq d$.

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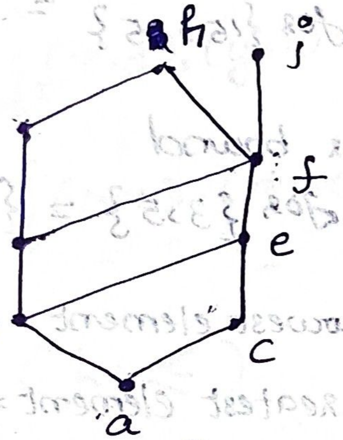
•) 

a) Find lower bounds and upper bounds of the subset $\{a,b,c\}$, $\{j,h\}$, $\{a,c,d,f\}$

ans:- upper bound = $\{e,f,h,j\}$
lower bound = $\{a\}$ — $\{a,b,c\}$

upper bound = Nil.
lower bounds = $\{f,e,c,a,b,d\}$ — $\{j,h\}$

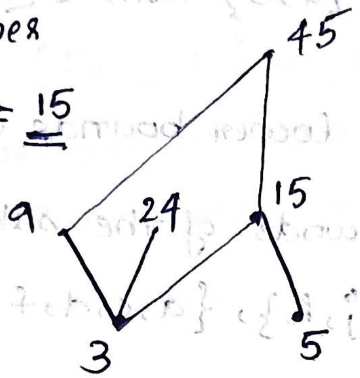
upper bound = $\{h,j\}$
lower bound = $\{a\}$ — $\{a,c,d,f\}$

•) For the poset $(\{3,5,9,15,24,45\}, |)$ [a divides b.] Find maximal & minimal elements, upperbound and lower bound ~~for~~ for ~~a) {15,45}~~ the following.

an) $R = \{(3,9),(3,3),(3,15),(3,24),(3,45),(5,5),$
$(5,15),(5,45),(9,9),(9,45),(15,15),$
$(15,45),(24,24),(45,45)\}$.

Least upper bound for $\{3,5\} = \underline{\underline{15}}$

Greatest lower bound for $\{15,45\} = \underline{\underline{15}}$



Maximal elements = 24, 45

Minimal elements = 3, 5

Lower ~~upper~~ bound for $\{15,45\} = \{3,5,15\}$

reflexive (15,15)

upper bound for $\{3,5\} = \{15,45\}$

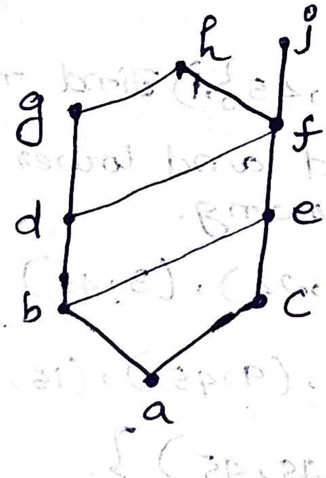
Lowest element = Nil.

Greatest element = Nil.

→ Least upper bound {LUB (supremum) and greatest → (join or sum) $[a \vee b,\ a \oplus b,\ a+b,\ a \cup b]$

→ lower bound: GLB (Infimum) → (meet or product) $[a \wedge b,\ a * b,\ a \cdot b,\ a \cap b]$

a) Find LUB & GLB of the subset $\{b, d, g\}$



$UB = \{g, h\}$

$LB = \{b, a\}$

$LUB = \{g\}$

$GLB = \{b\}$

• R' is defined as $aR'b$ if $a \not R b$

R' ↓ compliment $(\sim R)$

Eg:- $A = \{x, y, z\}$

$B = \{1, 2, 3\}$

Let R be a relation from $A \rightarrow B$, where $R = \{(x,1), (x,2), (y,3)\}$

$R' = ?$

ans) $R' = \{(x,3), (y,1), (y,2), (z,1), (z,2), (z,3)\}$ → Compliment

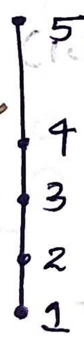
$R^{-1} = \{(1,x), (2,x), (3,y)\}$ → Inverse relation

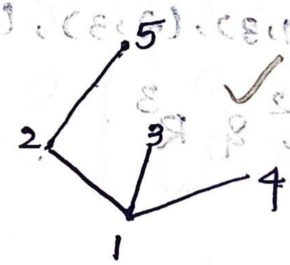
v.v.imp

- **Lattices $(L, \preceq)$**

A partially ordered set $(L, \preceq)$ in which every pair of elements has a least upper bound & a greatest lower bound is called a lattice.

Lattice





Poset but not lattice since all pairs does not have LUB & GLB

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imp

- Let R_1 & R_2 be a relations on a set A represented by matrices $M_{R_1} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ & $M_{R_2} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

Find matrices that represent

a) $R_1 \cup R_2$ b) $R_1 \cap R_2$ c) $R_2 \circ R_1$ d) $R_1 \circ R_1$

ans) a) $R_1 \cup R_2 = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

$\bar{R} \rightarrow$ transpose

$R^{-1} \rightarrow$ just x-1

$R^2 \rightarrow R \circ R$

binary addition

b) $R_1 \cap R_2 = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix}$

d) $R_1 \circ R_2$ $\cancel{R_2 \circ R_1}$ = ($M_{R_2} \times M_{R_1}$) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$

c) $R_2 \circ R_1 = M_{R_1} \times M_{R_2} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix}$

•) Let 'R' be the relation on the set $\{1, 2, 3, 4, 5\}$ containing the pairs $\{(1,1), (1,2), (1,3), (2,3), (2,4), (3,1), (3,4), (3,5), (5,4)\}$ find R^2 & R^3.

ans)

$$\begin{array}{c|c} & \begin{matrix} 1 & 2 & 3 & 4 & 5 \end{matrix} \\ \begin{matrix} 1 \\ 2 \\ 3 \\ 4 \\ 5 \end{matrix} & \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \end{array}$$

$$R^2 = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$R^3 = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

1) Let $S = \{-1, 0, 2, 4\}$ find $f(S)$ if

a) $f(x) = 1$

b) $f(x) = 2x+1$

c) $f(x) = \left\lceil \frac{x}{5} \right\rceil$ greater integer

d) $f(x) = \left\lfloor \frac{(x^2+1)}{3} \right\rfloor$ lowest integer

$(0, 0, 1, 5)$

ans:) a) $f(S) = 1$

b) $f(S) = \{-1, 1, 5, 9\}$

$2(-1)+1$, $2(0)+1$, $2(4)+1$

c) $f(S) = \{\cancel{0}\ 0, 1\}$

$0, 0, 1, 1$

$-\frac{1}{5} = -0.2 = 0$

$2/5 = 0.4 = 1$

$4/5 = 0.8 = 1$

Set no repetition

d) $f(S) = \{0, 1, 5\}$

Q) Let f be a function from $R \to R$ defined by $f(x) = x^2$,

find a) $f^{-1}(\{1\})$

b) $f^{-1}(\{x : 0 < x < 1\})$

c) $f^{-1}(\{x : x > 4\})$.

ans:) a) $\{-1, 1\}$

b) $(-1, 0) \cup (0, 1)$

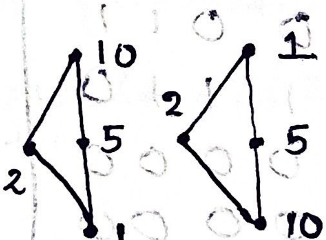
c) $(-\alpha, -2) \cup (2, \alpha)$

28/2/25 • **Principle of Duality**.

Suppose, $(S, \preceq)$ is a poset then $(S, \succeq) \to$ converse of that relation. is also a poset.

Eg:- $(S, \preceq) = (D_{10}, |)$ $\hookrightarrow$ divides

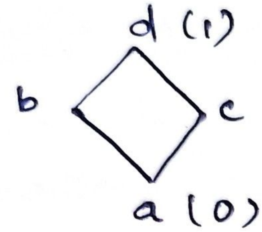
then $(S, \succeq) = (D_{10}, *)$



Incase of lattices if $(L, \preceq)$ $\hookrightarrow$ lattice is a lattice then, $(L, \succeq)$ is also a lattice also the operations of join and meet on $(L, \preceq)$ becomes the operations of meet and join respectively on $(L, \succeq)$.

Eg:- If L is the collection of 12 partitions of $S = \{1,2,3,4\}$ ordered such that $P_i \leq P_j$ if each block of P_i is a subset of a block P_j, show that ~~the~~ ~~t~~ L is a bounded lattice and draw its Hasse diagram.

Note:- Bounded lattice has an upperbound (1) & lower bound (0) represented as




→ not a bounded lattice since no UB & LB.

ans:- The 12 partitions of $S = \{1,2,3,4\}$ are

$P_1 = \{(1),(2),(3),(4)\}$ ie, $[1,2,3,4]$

$P_2 = \{(1,2),(3),(4)\}$ ie, $[12,3,4]$

$P_3 = [13,2,4]$ $\quad P_{10} = [134,2]$

$P_4 = [14,2,3]$ $\quad P_{11} = [234,1]$

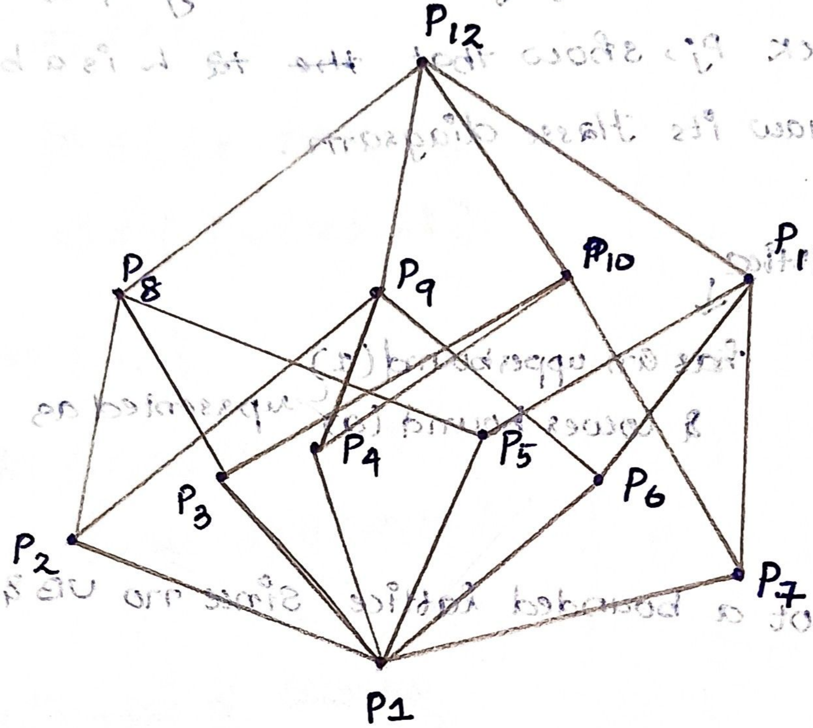
$P_5 = [23,1,4]$ $\quad P_{12} = [1234]$

$P_6 = [24,1,3]$

$P_7 = [34,1,2]$

$P_8 = [123,4]$

$P_9 = [124,3]$



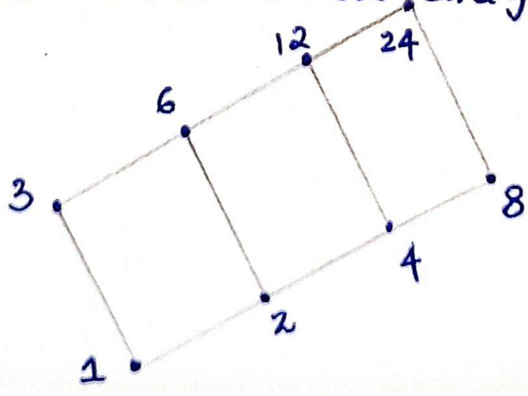
It is a bounded lattice [has UB & LB]

UB = P_{12} , LB = P_1

Eg:) If S_n is the set of all divisors of the positive integer n and D is the relation of 'division', viz., aDb if and only if a divides b, prove that $\{S_{24}, D\}$ is a lattice. Find also all the sublattices of D_{24} $[=\{S_{24}, D\}]$ that contain 5 or more elements.

ans) $\{S_{24}, D\} = \{(1, 2, 3, 4, 6, 8, 12, 24), D\}$

is a lattice whose Hasse diagram is



The sublattices containing 5 elements are {1, 2, 3, 6, 12}, {1, 2, 3, 12, 24}, {1, 2, 6, 12, 24}, {1, 3, 6, 12, 24} and {1, 2, 4, 8, 24}. The sublattice containing 6 elements is {1, 2, 3, 6, 12, 24}.

6/3/25

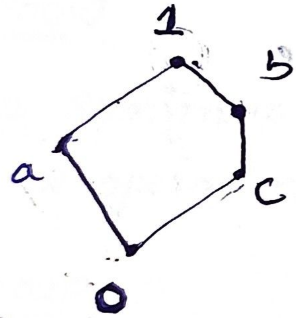
• Distributive lattice

LUB GLB

→ A lattice $(L, \vee, \wedge)$ is said to be distributive if for any element $a, b, c \in L$,

$$a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c) \text{ and}$$

$$a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$$

→ 

Check whether ~~the~~ g if it is distributive or not.

ans:-) $\cancel{b \vee c = b}$ LUB (i) $a \wedge (b \vee c) = (a \wedge b) \vee (a \wedge c)$

$\cancel{(a \wedge b)}$ $= a \wedge b =$

$\cancel{=}$ $=$

$=$

(ii) $a \vee (b \wedge c) = (a \vee b) \wedge (a \vee c)$

$a \vee c$

(iii) $c \wedge (a \vee b) = (c \wedge a) \vee (c \wedge b)$

$c \wedge 1$

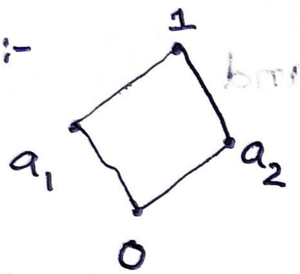
→ Compliment Lattice

If $(L, \wedge, \vee, 0, 1)$ is a bounded lattice and $a \in L$ then an element $b \in L$ is called a compliment of a if

$a \vee b = 1$ & $a \wedge b = 0$

If every element of L has atleast one compliment then the lattice is called complimented lattice.

Eg:-



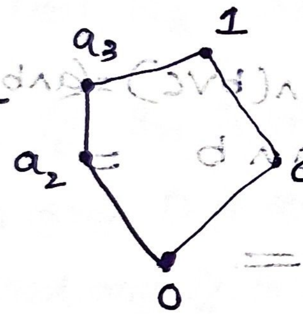
$a_1 \wedge a_2 = 0$
$a_1 \vee a_2 = 1$

$a_1' = a_2$ $\quad 1' = 0$ (opp elements)

$a_2' = a_1$ $\quad 0' = 1$

∴ complimented lattice

Eg:-



$a_1' = a_2, a_3$ $\quad 1' = 0$

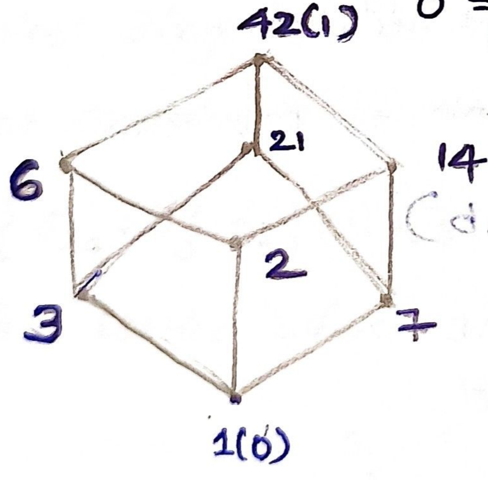
$a_2' = a_1$

$a_3' = a_1$

$0' = 1$

complimented lattice

Eg:-



mostly opp case

$3' = 14$ $\quad 1' = 42$

$42' = 1$ $\quad 14' = 3$

$7' = 6$ $\quad 21' = 2$

$6' = 7$ $\quad 2' = 21$

Complimented lattice

→ Algebraic system

A set together with a certain operations for combining the elements of the set to form other elements of the set is usually referred to as an algebraic system.

Semigroup

If 'S' is a non empty set and '*' be a binary operation on S then the algebraic system $\{S, *\}$ is called a semigroup if, the operation '*' is associative. i.e, for any element $a, b, c \in S$ $(a*b)*c = a*(b*c)$.

Monoid

If a semigroup $\{M, *\}$ has an identity element with respect to the operation '*', then $\{M, *\}$ is called a monoid.

Group

If 'G' is a non empty set and '*' is a binary operation of 'G' then the algebraic system $\{G, *\}$ is called a group if these conditions are satisfied.

(i) Associativity (Semigrp)
(ii) Existence of identity (monoid)
(iii) Existence of inverse

Abelian group / Commutative group

A group with commutative property

Example for Semigroup

Let 'E' is the set of positive even integers, then

7/3/25 $\{E, +\}$, $\{E, *\}$ are semigroup

$a*(b*c) = (a*b)*c$ $\qquad$ $a+(b+c) = (a+b)+c$

$2*(3*4) = (2*3)*4$ $\qquad$ $2+(3+4) = (2+3)+4$

$2*12 = 6*4$ $\qquad$ $2+7 = 5+4$

$\underline{\underline{24}} = \underline{\underline{24}}$ $\qquad$ $\underline{\underline{9}} = \underline{\underline{9}}$

Note:- Natural numbers include zero also

$\rightarrow$ identity = 0. $\qquad$ so not monoid

But $\{E, +\}$, $\{E, *\}$ are not monoid

identity = 1 $\rightarrow$ not a positive even integer

Eg:) If $*$ is the binary operation on the set R of real numbers defined by $a*b = a+b+2ab$ a) find if $\{R, *\}$ is a semigroup, is it commutative

b) find the identity element if exist

c) which elements have inverse and what are they?

ans) a) $a*(b*c) = (a*b)*c$; $a*b = a+b+2ab$

$a*(b*c) = \cancel{a+b+2ab}\ a*(b+c+2bc)$

$= a+b+c+2bc + 2a(b+c+2bc)$

$= \underline{\underline{a+b+c+2bc+2ab+2ac+4abc}}$

$(a*b)*c = (a+b+2ab)*c$

$= a+b+2ab+c + 2(a+b+2ab)(c)$

$= a+b+2ab+c+(2a+2b+4ab)c$

$= a+b+c+2ab+2ac+2bc+4abc$

$= a+b+c+2bc+2ab+2ac+4abc$. $\Rightarrow$ Semigroup

~~Commutative~~

b) Commutative.

$a*b = a+b+2ab$

$b*a = b+a+2ba$ $\quad a*b = b*a \Rightarrow$ Commutative.

b) $a*e = a \rightarrow$ identity

$a+e+2ae = a$

$e+2ae = a-a = 0$

$e+2ae = 0$

$e(1+2a) = 0$

$\therefore e = 0$ //

or & $1+2a = 0$

$2a = -1$

$a = -1/2$.

identity element $= 0$ (e)

c) $a^{-1} * a = e \leftarrow$ inverse. $\quad e = 0$

$a + a^{-1} + 2aa^{-1} = 0$

$$a^{-1} = \frac{-a}{1+2a}.$$

rational no.

Eg:- If $*$ is the operation defined on S, $S = Q \times Q$, where set of ordered pairs of rational numbers and given by $(a,b) * (x,y) = (ax, ay+b)$.

a) Find if $\{S, *\}$ is a semigroup, is it commutative.

b) Find the identity element of S.

c) which element if any, have inverse & what are they?

an) a) $a*(b*c) = (a*b)*c$

~~$a*(b*c) = (a, a*$~~

$(a,b) * ((x,y) * (p,q)) = (a,b) * (xp, xq+y)$

$= (axp, a(xq+y)+b)$

$= (axp, axq + ay + b)$

$(a*b)*c = [[(a,b) * (x,y)] * (p,q)]$

$= [(ax, ay+b) * (p,q)]$

$= axp, axq + ay + b$

$\Rightarrow$ Semigroup.

Commutative.

$(a,b)*(x,y) = (ax, ay+b)$

$\Rightarrow$ not commutative

$(x,y) * (a,b) = (ax, xb+y)$

b) $(a,b) * (e_1, e_2) = (a,b) \rightarrow$ identity

$(ae_1, ae_2 + b) = (a,b)$

Identity element $= (e_1, e_2) = (1,0)$

$ae_1 = a \qquad ae_2 + b = b$

$e_1 = a/a \qquad ae_2 = 0$

$= 1 \qquad e_2 = 0$

$a * a^{-1} = e$

c) $(a,b) * (a^{-1}, b^{-1}) = (1,0)$

$(aa^{-1}, ab^{-1} + b) = (1,0)$

$aa^{-1} = 1 \qquad ab^{-1} + b = 0$

$a^{-1} = 1/a \qquad ab^{-1} = -b$

$b^{-1} = -b/a.$

Eg:) Show that set Q^+ form an abelian group under the operation $*$ defined by $a*b = \frac{1}{2}ab$ when $a, b \in Q^+$

an) abelian group $\rightarrow$ associative $(a*b)*c = a*(b*c)$

Identity $(a*e = a)$

Inverse $(a^{-1}*a = e)$

Commutative $(a*b = b*a)$

(i) $(a*b)*c) = a*(b*c)$

$(\frac{1}{2}ab * c) = a* (\frac{1}{2}bc) \rightarrow$ associative.

$\underline{\underline{\frac{1}{4}abc}} = \underline{\underline{\frac{1}{4}abc}}$ $\Rightarrow$ semigroup

(ii) $a*e = a$ $\qquad \underline{\underline{e=2}}$

$\frac{1}{2}ae = a$ $\qquad$ Identity element $= 2$.

$ae = 2a$ $\qquad \Rightarrow$ Monoid.

$e = 2a/a = \underline{\underline{2}}$

(iii) $a^{-1}*a = e$

$\frac{1}{2}a^{-1}a = 2$ $\qquad \Rightarrow$ Group

$a^{-1}a = 4$

$a^{-1} = \underline{\underline{4/a}}.$

(iv) $a*b = \frac{1}{2}ab$ $\qquad \Rightarrow$ Abelian group

$b*a = \frac{1}{2}ba$ $\qquad$ Commutative :

Ring:

An algebraic system $\{R, +, \cdot\}$, where R is a non-empty set and $+$ & $\cdot$ are two closed binary operations is called a ring if the following conditions are satisfied:-

(i) $\{R, +\}$ is an abelian group

(ii) $\{R, \cdot\}$ is a semigroup

(iii) The operation $\cdot$ is distributive over $+$

i.e., $a \cdot (b+c) = a \cdot b + a \cdot c$ and

$(b+c) \cdot a = b \cdot a + c \cdot a$

10/3/25

→ If $(R, \cdot)$ is commutative then the ring $(R, +, \cdot)$ is called a commutative ring

→ If $(R, \cdot)$ is a monoid then the ring $(R, +, \cdot)$ is called a ring with identity or unity.

→ If a, b are two non-zero elements of a ring R such that $a \cdot b = 0$, then a & b are divisors of zero or zero divisors.

→ A commutative ring with unity and without zero divisors is called an integral domain.

→ Let R be a ring with unity 'U'. If $a \in R$ and there exist $b \in R$ such that $(a, b) = (b, a) = U$ then, b is called multiplicative inverse of a and a is called unit of R [The element b is also a unit of R]

1. Find the additive inverse for each element in the rings of examples 13.5 and 13.6.

Ans) 13.5

For $R = \{a, b, c, d, e\}$ we define $+$ and $\cdot$ by Tables 13.2(a) and (b).

Table 13.2(a)

$+$	a	b	c	d	e
a	a	b	c	d	e
b	b	c	d	e	a
c	c	d	e	a	b
d	d	e	a	b	c
e	e	a	b	c	d

$\cdot$	a	b	c	d	e
a	a	a	a	a	a
b	a	b	c	d	e
c	a	c	e	b	d
d	a	d	b	e	c
e	a	e	d	c	b